INCH-POUND

MIL-DTL-45444C
W/AMENDMENT 2
23 November 2020
SUPERSEDING
MIL-DTL-45444C
W/AMENDMENT 1
17 April 1999

DETAIL SPECIFICATION

CYCLOTETRAMETHYLENETETRANITRAMINE (HMX)

Reactivated after23 November 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for two grades of HMX for use in explosive compositions as follows:

Grade A - 93 percent minimum purity with the remainder being RDX
Grade B - 98 percent minimum purity with the remainder being RDX

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: US Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or email to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

AOP-7	-	Manual of Data Requirements and Tests for the Qualification of Explosive Materials for Military Use
STANAG 4515	-	Explosives: Thermal Characterization by Differential Thermal Analysis, Differential Scanning Calorimetry and Thermogravimetric Analysis
STANAG 4489	-	Explosives, Impact Sensitivity Tests

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

CCDC – AC DRAWINGS

12982865	-	Minimum Marking Instructions for Ammunition and Explosive Containers

Holston Army Ammunition Plant/Holston Defense Corp. Engineering Drawing No. 7651-1028.48, 7651-1028.49 & 7651-1028.50 - Holston Impact Machine

Naval Sea Systems Command, Code Ident 10001, Drawing No: LD70518 - Impact Machine

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Wire Cloth and Sieves for Testing Purposes
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org.)

ANSI/ISO/ASQC A8402-1994 - Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Ave, Milwaukee, WI 53202)

2.4 Order of precedence. Unless otherwise specified in this document or in the contract, in the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.4).

3.2 Materials. HMX shall conform to the chemical and physical requirements specified in Table I, when tested as specified in the applicable paragraphs (See also 4.6, 6.10 and 6.12).

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TABLE I. Chemical and physical requirements for HMX.

TEST	GRADE A	GRADE B	TEST PARAGRAPH
Alpha HMX, wt.% (max)	0.10	0.10	4.7.1
HMX content (Beta), wt % min	93	98	4.7.1
RDX content, wt.% (max)	7	2	4.7.1
Melting Point, Celsius (deg. C), min	277	277	4.7.2
No. of Insoluble particles, on USS No. 40 sieve (max) on USS No. 60 sieve (max)	 0 5	 0 5	4.7.3
Acetone insoluble material, wt.% (max)	0.05	0.05	4.7.4
Inorganic insoluble, wt.% (max)	0.03	0.03	4.7.5
Acidity, wt.% (max)	0.02	0.02	4.7.6
Impact sensitivity, cm (min), (See note 1) ERL, type 12, 2.5 kg Holston impact method Short method (for acceptance only) Long Method	 17 26 26	 17 26 26	 4.7.7.3 4.7.7.4.1 4.7.7.4.2
Granulation	See Table II	See Table II	4.7.8
Workmanship	See 3.4	See 3.4	4.7.9

Note 1 – Either the ERL, type 12, 2.5 kg impact sensitivity test method or the Holston impact sensitivity test method may be used. The ERL impact test method is described in AOP-7. The ERL, type 12 machine is shown in the Naval Sea System Command impact machine drawing (see 2.2.2). The Holston impact machine is shown in the Holston AAP impact machine drawing (see 2.2.2).

3.2.1 Granulation. HMX shall be designated as Class 1 to Class 6 in compliance with the granulation requirements given in Table II, when tested in accordance with 4.7.8.

TABLE II. Granulation.

Through U.S. Standard Sieve No.	Class 1 Wt. %	Class 2 Wt. %	Class 3 Wt. %	Class 4 Wt. %	Class 5 Wt. %	Class 6 Wt. %
8				100		
12			99 min	85 min		99 min
35				25 ± 15		
50	90 ± 6	100	40 ± 15			90 min
100	50 ± 10		20 ± 10	15 max		65 ± 15
120		98 min				
200	20 ± 6		10 ± 10			30 ± 15
325	8 ± 5	75 min			98 min	15 ± 10

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3.2.2 Use of virgin HMX only. HMX covered by this specification shall consist of virgin HMX only (see 6.7).

3.3 Process controls. The contractor shall submit a Process Control document to the Government specifying the process variables which are considered crucial for the production of HMX. The Process Control Document shall be submitted 30 days prior to the commencement of production in accordance with 4.2.

3.4 Workmanship. The manufacturer shall use procedures and controls which assure that the HMX produced does not contain foreign material such as dirt, rust, paint or metal chips, etc., and that the safety and reliability of the explosive are not compromised. Compliance with this requirement shall be as specified in 4.7.9.

3.5 Product certification. No HMX material whether produced for military or commercial use may be represented as meeting the requirements of this detail specification unless it complies with all requirements contained in Section 3. The process control document (see 3.3), test data for first article inspection (see 4.4) and conformance inspection (see 4.5) shall be submitted to the Technical Agency (see 6.9) for evaluation, approval and certification on compliance with MIL-DTL-45444.

4. VERIFICATION

4.1 General provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Reference shall be made to ANSI/ISO/ASQC A8402-1994 in order to define the terms used herein. All items must meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted (see 6.3):

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications, shall be available upon request by the contracting officer.

d. Number of pounds of HMX in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with the subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Process control document. A Process Control Document shall be submitted to the Technical Agency (see 6.9). The document shall contain a description of the process, all materials used, process conditions, procedures, production and inspection equipment used to produce HMX meeting the requirements of this specification. In addition, whenever there is a change that requires lot interfix change (per MIL-STD-1168) such as the source of materials, process conditions, procedures, etc., from those in the previous process control document, the Technical Agency shall be notified of the new process and process control change(s) and the reason for the change(s) within 10 business days from the change(s).

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (See 4.4)

b. Conformance inspection (See 4.5)

4.3.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2, (see 6.2c). The first article shall consist of five kilograms (11 pounds) of the composition obtained by sampling as described in 4.6.1. The samples shall be obtained from a production batch which has been produced by the contractor using the same production process, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production. All samples submitted shall have been produced by the contractor using the same production process, procedures, equipment as will be used in regular production, and shall be accompanied by certificates of analysis. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample

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containing known defects will not be submitted unless specifically authorized by the Contracting Officer, (see 6.2).

4.4.2 Inspections to be performed. See MIL-STD-1916 for definitions of critical, major and minor classification of characteristics.

4.4.3 Rejection. If any sample fails to comply with any of the applicable requirements (see 3.), the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the requirements.

4.5 Conformance inspection.

4.5.1 Inspection lot formation. A lot shall consist of one or more batches of HMX produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of HMX that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (homogeneity criteria). Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. The lot shall be submitted for inspection in accordance with MIL-STD-1916 when applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.5.2 Examinations and tests.

a. Classification characteristics. Conformance examinations and tests are specified in the following classification of characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited in the specification (4.7.9.1 to 4.7.9.5), attributes sample inspection shall be conducted in accordance with Table III below, using the inspection levels stated in the classification of characteristics paragraphs. (See MIL-STD-1916 for definitions of critical, major and minor classification of characteristics.)

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TABLE III. Attributes sample inspection.

Lot size	Inspection Levels		
	I	II	III
2 to 8	*	*	*
9 to 15	*	*	*
16 to 25	*	*	*
26 to 50	*	*	32
51 to 90	*	*	32
91 to 150	*	125	32
151 to 280	*	125	32
281 to 500	*	125	32
501 to 1200	*	125	80
1201 to 3200	*	125	80
3201 to 10000	1250	125	125
10001 to 35000	1250	315	125
35001 to 150000	1250	315	125
150001 to 500000	1250	500	200
500001 and over	1250	500	200

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative conformance inspection provisions. Unless otherwise specified herein or provided for in the contract, alternative conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of verification required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.11). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified conformance inspection provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative conformance inspection provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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4.5.2.1 Classification of characteristics - HMX.

Classification	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Alpha HMX content	4.6	3.2	4.7.1
102	HMX content	4.6	3.2	4.7.1
103	RDX content	4.6	3.2	4.7.1
104	Melting point	4.6	3.2	4.7.2
105	Insoluble particle	4.6	3.2	4.7.3
106	Acetone insoluble material	4.6	3.2	4.7.4
107	Inorganic insoluble	4.6	3.2	4.7.5
108	Acidity	4.6	3.2	4.7.6
109	Impact sensitivity	4.6	3.2	4.7.7
110	Granulation	4.6	3.2.1	4.7.8
111	Workmanship	4.6	3.4	4.7.9
Minor	None defined			

4.6 Sampling and inspection equipment.

Hazard Notice

The materials described herein are flammable or explosive or both. Consequently, they present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically-induced incidents.

4.6.1 Sampling. A representative sample of approximately 300 grams shall be taken from each batch of HMX in accordance with ASTM Procedure E300 for solids. The selection of batches for inspection sampling and testing shall be in accordance with MIL-STD-1916 except that in lieu of Table IV (MIL-STD-1916), the following continuous sampling plan shall be used: i=10, f=1/3 and there shall be no reduced or tightened inspection level. (AQL are provided as indices to simplify the use of the table in MIL-STD-1916, but have no other meaning relative to the sampling plan.) The samples shall be tested for compliance with requirements of 3.2, Table I of this specification. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.7. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch the contractor will return to 100 % inspection until "i" successive batches are accepted as required by MIL-STD-1916.

4.6.2 Inspection equipment. Commercially available inspection equipment shall be used for all examinations and tests specified in 4.5 and 4.7. The contractor is responsible to have the

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necessary equipment available, use it correctly and to assure that proper calibration procedures are followed. The designation of an instrument of a particular make or model number is not intended to preclude the use of an equivalent make or model which is capable of performing the required test. Government approval of all inspection equipment is required prior to its use for acceptance purposes. Inspection equipment designs shall be submitted to the Technical Agency (see 6.9).

4.7 Methods of inspection. The following tests shall be performed using prescribed analytical procedures for replicate determination given in standard and analytical textbooks. Also, unless otherwise specified herein, all chemicals and reagents shall be ACS Grade or reagent grade chemicals (see 6.10 for use of equivalent test methods).

4.7.1 Purity of HMX. The alpha HMX, beta HMX and RDX content of HMX shall be determined by x-ray diffraction as follows:

Instrumentation, Equipment and Supplies

a. Commercially available powder X-Ray Diffractometer/XRD (e.g., PANalytical (Philips), Bruker, Rigaku) consisting of:
1. High voltage generator.
2. Copper (Cu) target X-Ray tube.
3. Sample stage/platform.
4. Wavelength filtering system for removing Cu K-β (beta) radiation.
5. Wavelength focusing system for Cu K-α (alpha) radiation.
6. Detector such as a scintillation, Peltier or transducer type.
7. Goniometer or similar system for precise control of source, sample and detector angular positions.
8. Computer and software for diffractometer control, sample/standard testing, results analysis and calculation.

b. Sample press capable of consistent ram loads of approximately 100lbs.
c. Laboratory safety shields for grinding and pressing of samples.
d. Stable tube decay standard such as National Institute of Standards and Technology (NIST) Standard Reference Material (SRM) 640 Silicon Wafer abbreviated as Si640.
e. Stable Instrument Response Standard such as NIST SRM1976 Alumina wafer abbreviated as Al1976.
f. Additional equipment necessary for sample preparation according to XRD manufacturer's instructions (e.g., sample holders, spatula, press block, razor blade or similar straight edge).
g. High purity (min. of 99.90%) alpha (α) HMX, beta (β) HMX and RDX to be used for calibration standards. Refer to Section 6.8 for sources of standard material.

4.7.1.1 Daily (each day of XRD use) analysis of tube decay and intensity standards.

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X-ray tubes decay with use therefore the intensities will decrease over time. X-ray Tubes should be replaced when they burn out or the intensity standard runs outside of 3 SD control limits established as outlined in the intensity standard section below.

A. Tube Decay Standard (Si640).

The above necessitates that a stable standard such as a NIST SRM640 silicon wafer (Si640) be measured daily to calculate an intensity correction factor. Si640 has a characteristic peak approx. 28.4°2Θ, see Figure 1. The following steps may be used as a guide:

a. When a new tube is installed analyze the Si640 standard at least 20 times using a scan program that provides suitable peak resolution while maximizing peak cps between 28.0°2Θ and 30.0°2Θ. Calculate the average intensity in cps. This is the baseline SI640 (cps) intensity for the new tube.

b. Each day of use thereafter the daily Si640 (cps) is measured to determine the daily intensity correction factor.

$$\text{Daily Intensity Correction Factor} = \frac{1}{\left(\frac{\text{Daily Si640 (cps)}}{\text{Baseline SI640 (cps)}}\right)}$$

c. For this day all sample and/or standard corrected cps are multiplied by this daily factor.

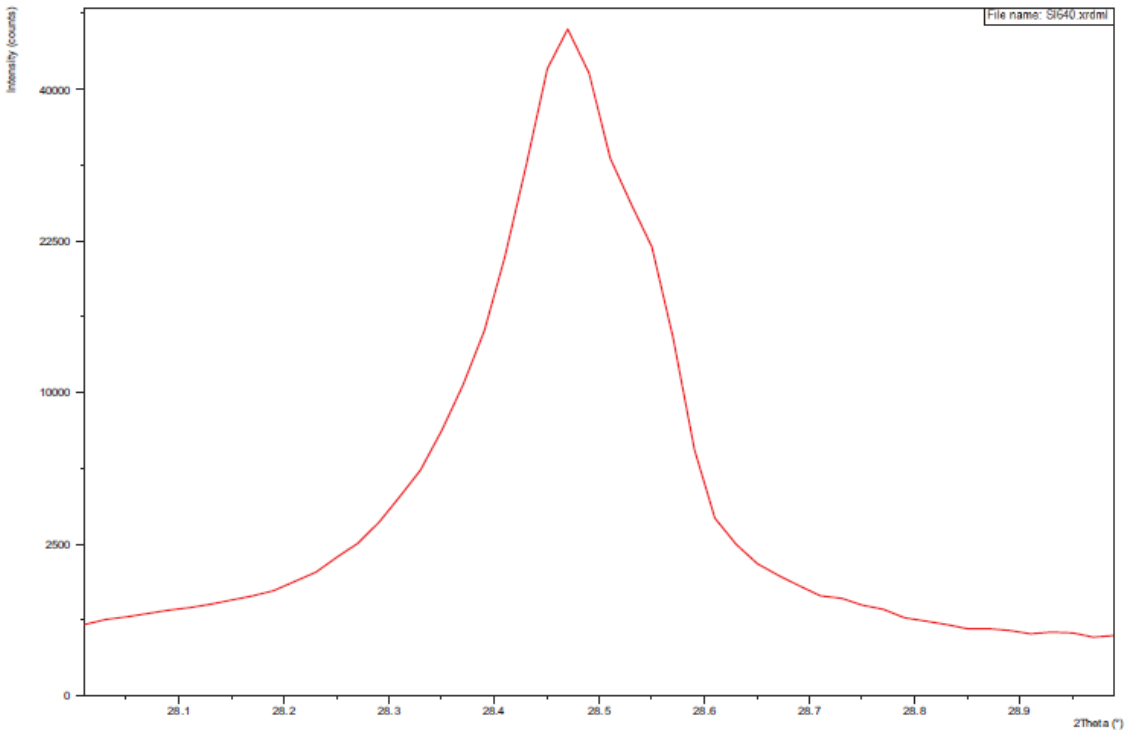


Figure 1. Si640 characteristic peak at approx. 28.4°2Θ.

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B. Intensity Standard (Al1976).

A daily intensity standard such as a NIST SRM1976 alumina wafer (Al1976) is suggested to verify instrument alignment and performance. Al1976 has a characteristic peak at approx. 25.6°2Θ, Figure 2. The following steps may be used as a guide:

a. When a new tube is installed analyze the Al1976 standard at least 20 times using a scan program that provides suitable peak resolution while maximizing peak cps between 25.0°2Θ and 26.0°2Θ. Calculate the average intensity in cps and the standard deviation (SD). The acceptable limits for this standard shall be the average cps ±3 SD.

b. Each day of use thereafter Al1976 is measured and the corrected cps calculated. This value must fall within the limits determined when the new tube was installed.

$$\text{Corrected Alumina cps} = \text{Measured Alumina cps} \times \frac{\text{Daily Factor}}{\text{Baseline Factor}}$$

Where

Daily Factor = calculated as outlined in Tube Decay standard section above
Baseline Factor = Daily Factor at time intensity standard control limits were established

c. If the cps is not within limits samples cannot be analyzed until the cause is determined and corrected. This is also true if a peak shift occurs (> 0.5°2θ).

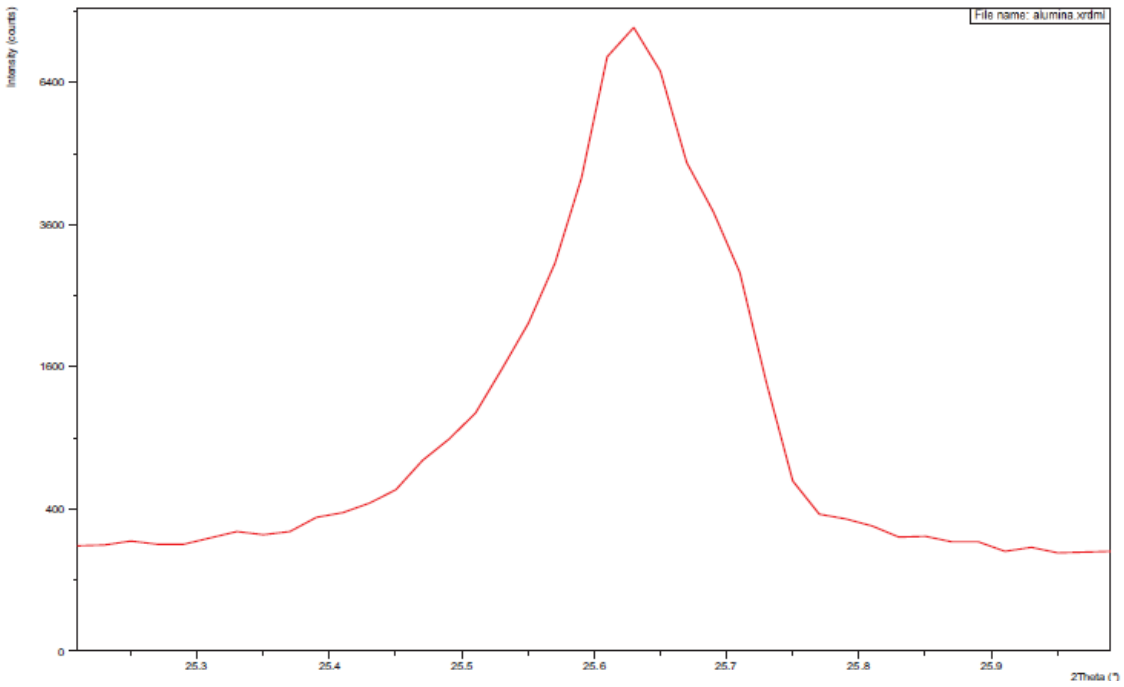


Figure 2. Al1976 Characteristic Peak at approx. 25.6°2Θ.

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4.7.1.2 Preparation of calibration standard samples.

Using a mortar and pestle, grind the individual standards behind a safety shield until all powder passes an ASTM E11 USS 230 Sieve (63 µm). Prepare the following % w/w homogenous standard mixtures in quantities sufficient to fill the sample holders used:

B HMX, %	α HMX, %	RDX, %
99.70	0.30	0.00
99.40	0.60	0.00
99.00	1.00	0.00
98.00	2.00	0.00
97.00	3.00	0.00
96.00	4.00	0.00
95.00	5.00	0.00
99.00	0.00	1.00
98.00	0.00	2.00
97.00	0.00	3.00
96.00	0.00	4.00
95.00	0.00	5.00
94.00	0.00	6.00
93.00	0.00	7.00
92.00	0.00	8.00
91.00	0.00	9.00
90.00	0.00	10.00

4.7.1.3 Analysis of calibration standards and construction of calibration and correction curves.

a. Using a safety shield and the XRD manufacturer's instructions press each standard into a sample holder. Pressed sample surfaces must be smooth, flat and not have any voids, cracks, ridges, etc. A force of approx. 100 lbs. has been found to be sufficient.

b. Load standards onto XRD and operate in accordance with the manufacturer's instructions.

c. Optimal tube voltage and filament current can be selected by analyzing five of the calibration standards at various settings. An example setting would be 45Kv and 20mA although these will vary depending on the XRD used and system settings.

d. Use a scan program that provides suitable peak resolution and shape while maximizing counts per second (cps) at the areas of interest. Example scan program:

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Starting Angle (degrees 2 theta (°2Θ))	16.0
End Angle (°2Θ)	26.0
Step Size (°2Θ)	0.03
Time per Step (s)	5
Scan Speed (°2Θ/s)	0.006
No. of Steps (#)	333

e. Determine the counts per second (cps) for the standards containing α HMX at ~25.10°2Θ and at ~17.81°2Θ for those containing RDX. These angles will vary according to the alignment of the instrument. Determine background cps for α HMX at 24.10°2Θ and for RDX at 16.90°2Θ.

f. Determine the corrected intensities for each standard by subtracting the background cps from the component counts as such:

Corrected cps for αHMX = (cps at approx. 25.10° - cps at 24.10°)
Corrected cps for RDX = (cps at approx. 17.81º - cps at 16.90º)

g. Using an appropriate software program (e.g., MS Excel) plot the corrected cps (Y) versus the standard concentrations (X). Draw the best fit line and calculate the linear regression equation
y = mx + b where:

y = cps
m = slope
x = concentration
b = y-intercept

The correlation coefficient, r^2, shall be ≥ 0.99.

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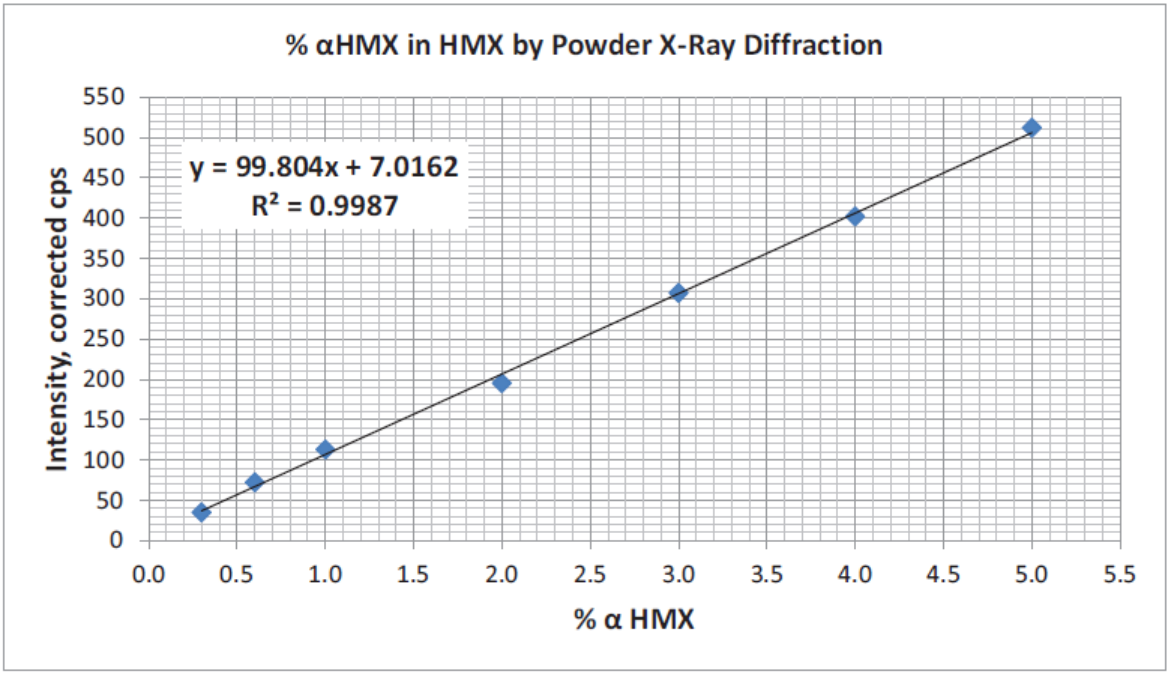


Figure 3. Example α HMX calibration curve.

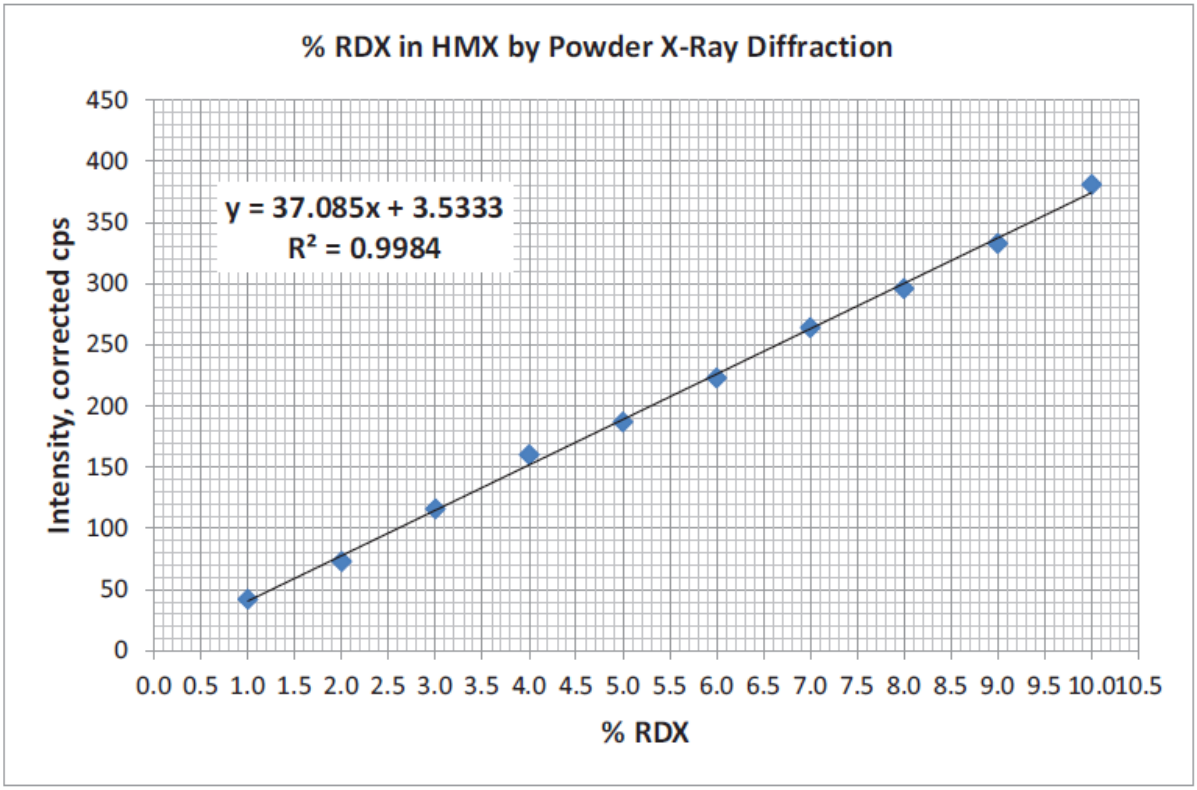


Figure 4. Example RDX calibration curve.

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h. RDX has a peak that overlaps the measurement peak of αHMX as shown in Figure 5. When present at >1.0%, this RDX peak interferes with αHMX peak. Therefore the resulting corrected αHMX cps requires further correction. This is done by using the linear formula determined from the RDX calibration curve (example shown in Figure 6). This curve is generated by measuring the maximum peak intensities at approx. 25.10º generated from those standards containing only RDX. The below formula calculates the attributable intensity (cps) that is due strictly to RDX. This intensity (cps) must be subtracted from the measured alpha intensity to correct for presence of RDX in the sample.

$y = mx + b$

Where: y = RDX correction cps
m = Slope of the line
x = % RDX
b = y-intercept

Corrected αHMX (cps) = measured αHMX (cps) – RDX correction (cps)

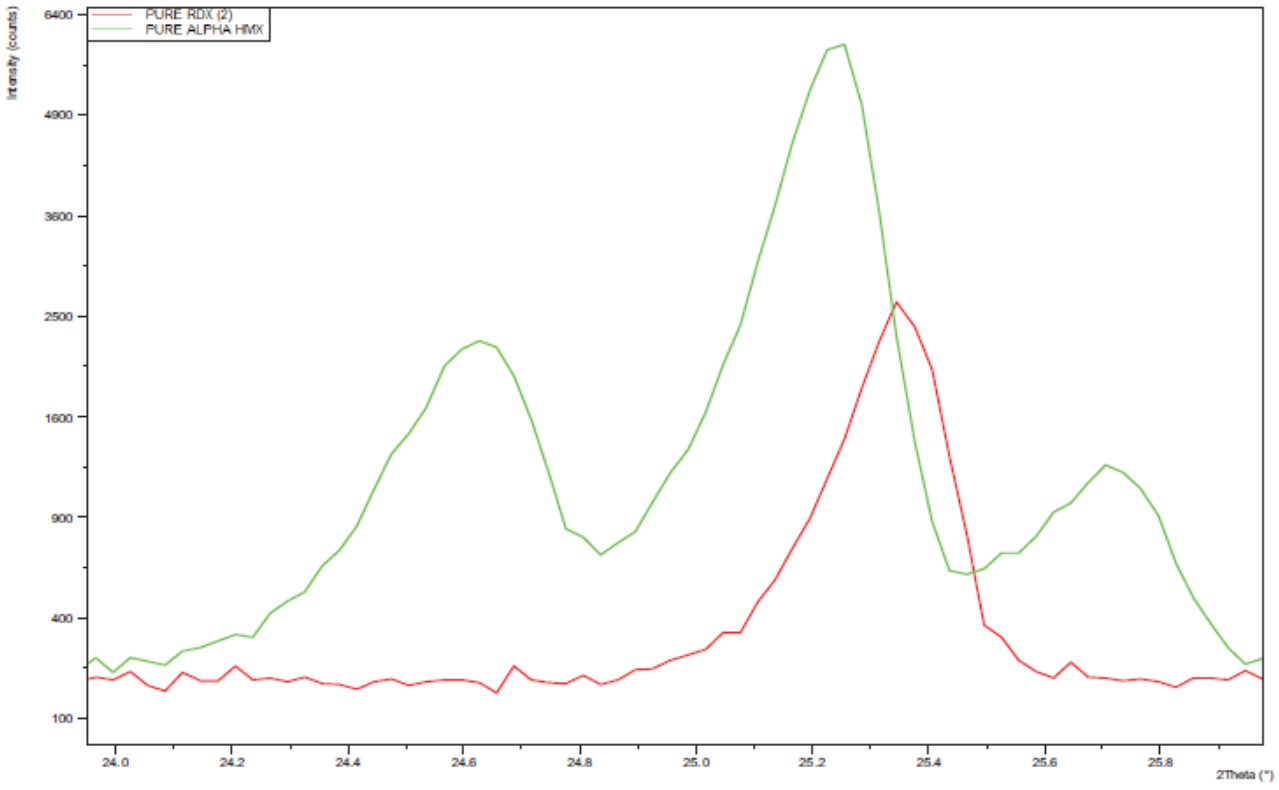


Figure 5. Example of RDX and αHMX peak overlap.

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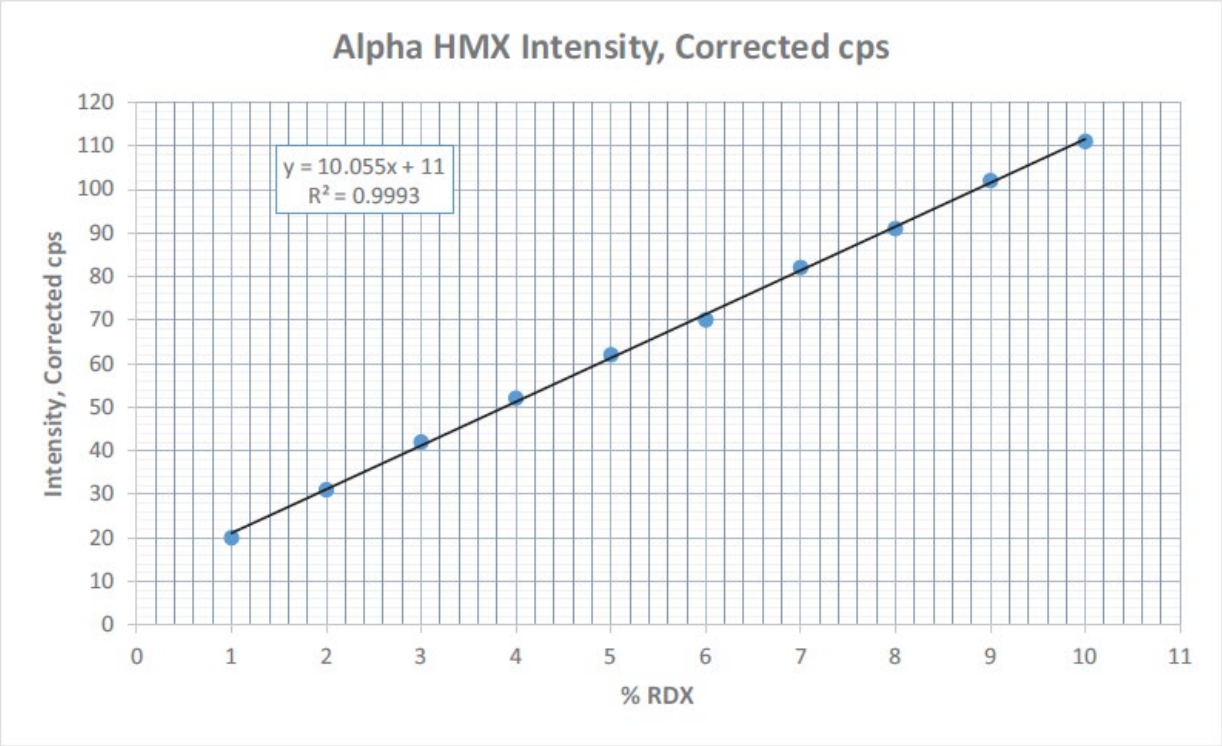


Figure 6. Example α HMX correction curve in the presence of RDX.

4.7.1.4 Preparation and analysis of HMX samples.

a. Using a safety shield and the XRD manufacturer's instructions press each sample into a sample holder. Pressed sample surfaces must be smooth, flat and not have any voids, cracks, ridges, etc. It may be necessary to reduce the particle size of the HMX by gently grinding small quantities with a mortar and pestle behind a safety shield to meet these requirements. A force of approx. 100 lbs. has been found to be sufficient for pressing samples.

b. Load samples onto XRD and operate in accordance with the manufacturer's instructions. Use the same scan program that was used for the calibration standards.

c. Record cps for each sample at the defined background and measurement degrees for RDX and αHMX.

4.7.1.5 Calculation of results.

a. Calculate the samples corrected cps by subtracting the background cps from the measurement cps for both RDX and αHMX.

Corrected cps for αHMX = (cps at approx. 25.10° - cps at 24.10°)
Corrected cps for RDX = (cps at approx. 17.81° - cps at 16.90°)

b. Enter sample data into a suitable software program (e.g., MS Excel) to calculate the %RDX and % αHMX using the linear regression equations determined for the standards. If the

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%RDX is >1.0% the αHMX cps for the sample require further correction before calculation of the % αHMX. βHMX is calculated by difference.

4.7.1.6 Alternate method for HMX purity and RDX content. A high performance liquid chromatographic (HPLC) method shall be used to determine HMX purity by quantifying the concentration of RDX impurity in the HMX product.

4.7.1.6.1 HPLC method conditions. The HPLC system used in the analysis of HMX product and appropriate method calibration standards shall be comprised of a column, mobile phase, pumping system, sample injection system, detector system and data capture system. The column used in the testing shall be a microBondapak CN (10 micron particle size, 3.9 mm x 150 mm Internal Dimensions; see 6.6.1) or a direct column equivalent. The mobile phase shall be a quaternary system consisting of the following components and volumetric proportions: 70% of 2,2,4-trimethylpentane (iso-octane), 20% of chloroform, 7% of acetonitrile and 3% of methanol. In actual application, the volumetric ratio of solvents can be adjusted slightly to "optimize" the chromatography of the components of interest (RDX and HMX). All solvents used in the mobile phase shall be vendor certified as HPLC grade chemicals. Prior to use in testing, the mobile phase should be thoroughly degassed using either vacuum degassing techniques or by purging with an inert gas. The injector component within the HPLC system shall be equipped with a sample loop capable of delivering a 20 microliter injection volume of sample to the HPLC column. The detection system used in this HPLC testing shall utilize an ultraviolet (UV) detector having a light wavelength of 254 nanometers.

4.7.1.6.2 Analysis of HMX samples and standards. Prior to the analysis of either HMX samples or standards, the pumping of mobile phase through the system shall be initiated at a flow rate of 1.5 milliliters per minute. The HPLC system shall be allowed to equilibrate at ambient conditions for approximately 10 minutes or until the flow of mobile phase and the output reading on the UV detector are stable. After the HPLC system has stabilized, the sample loop on the HPLC injector system shall be filled with an aliquot of the HMX sample or standard. Using either manual or automated injection techniques, the aliquot of sample in the 20 microliter sample loop on the injector should be transferred to the HPLC column. The peak area data from the HPLC chromatogram for both the HMX and RDX components in the samples and standards should be calculated and retained using the data collection system (i.e. computer, computing integrator, etc.).

4.7.1.6.3 Preparation and analysis of HMX calibration standards. Calibration standards having the approximate concentrations provided in Table V shall be prepared. The RDX and HMX used in the calibration standards shall have a minimum purity of 99.9%. Prepare the calibration standards by weighing the HMX and RDX components to the nearest 0.1 milligram in tared 250-mL glass volumetric flasks with ground glass stoppers. Add approximately 100 milliliters of HPLC grade acetone to the flasks. Cap the flasks and place the standards in an ultrasonic bath for 10 minutes or until the RDX and HMX are completely dissolved and uniformly mixed. Allow the standard mixtures to cool to ambient temperature, and then fill to the mark of the volumetric flasks using HPLC grade acetone.

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TABLE V. Composition of HMX/RDX Standards.

HMX		RDX	
Percent	Weight, grams	Percent	Weight, grams
99.5	0.4975	0.5	0.0025
98.0	0.4900	2.0	0.0100
96.5	0.4825	3.5	0.0175

Each calibration standard which is prepared shall be tested using the conditions and methodology outlined in Sections 4.7.1.6.1 and 4.7.1.6.2.

4.7.1.6.4 Preparation and analysis of HMX samples. Dry approximately 1 gram of the HMX sample to be tested in an oven at 100 ± 5˚C for a minimum of one hour. Using an analytical balance, weigh approximately 0.05 gram of the dried HMX to the nearest 0.1 milligram in a clean, glass 30-mL bottle with a Teflon lined lid. Add 25.0 milliliters of HPLC grade acetone to the bottle using an automatic repipette. Cap the bottle and place the samples in an ultrasonic bath for 10 minutes or until the HMX is completely dissolved and the solution is uniformly mixed.

The HMX samples which are prepared shall be tested using the conditions and methodology outlined in 4.7.1.6.1 and 4.7.1.6.2. Use the chromatographic peak areas for HMX and RDX in the sample to determine the HMX purity and the RDX content using the following calculations:

$$\text{Percent (HMX)} = \frac{100 \times A \times B}{C \times W}$$

Where:

A = The HMX chromatographic peak area (or height) obtained for the HMX sample.
B = The weight of HMX in the calibration standard in grams. (Use the calibration standard which provided a peak area closest to the area obtained for the sample.)
C = The HMX peak area obtained for the calibration standard having the weight, B. above.
W = The total weight of the HMX sample in grams.

$$\text{Percent (RDX)} = \frac{100 \times E \times F}{G \times W}$$

Where:

E = The RDX chromatographic peak area (or height) obtained for the RDX in the HMX sample.
F = The weight of RDX in the calibration standard in grams. (Use the calibration standard which provided a peak area closest to the area obtained for the sample.)
G = The RDX peak area obtained for the calibration standard having the weight, F. above.
W = The total weight of the HMX sample in grams.

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In the development of a calibration curve using the three HMX/RDX standards, upon establishing the linearity of the HPLC method response across the concentration range being studied, a single-point calibration using the standard which provided a peak area closest to the area obtained for the sample can be used to determine the HMX and RDX concentrations in the HMX product.

4.7.2 Melting point. Melting point of the HMX shall be determined using one of the following methods.

4.7.2.1 Fisher-Johns Melting point. A Fisher-Johns or equivalent hot state melting point apparatus shall be used to determine the melting point of HMX. The thermometer of this apparatus shall be calibrated using appropriate melting point standards. A portion of the sample shall be ground in small agate mortar and a very small quantity (approximately 0.05 gram) of the finely pulverized sample placed between two clean 18 millimeters (mm) diameter cover glasses. These shall be gently but firmly pressed together and placed in the circular depression on the stage. The powerstat shall be turned up and the unit allowed to heat. The heating rate may be very rapid to within 15 degrees C of the melting point. Thereafter a heating rate of approximately 1 degree C per minute shall be used. When the sample begins to melt, the thermometer shall be read and the temperature reading adjusted to reflect the thermometer calibration correction shall be reported as the melting point.

4.7.2.2 DSC Melting point (Alternate method). A Differential Scanning Calorimeter in accordance with STANAG 4515 using Nitrogen purge gas will be used which is capable of meeting the heating rate and sample size parameters as specified below.

a. Place 0.5 ± 0.25 mg of the HMX sample in an aluminum hermetic pan.
b. Place an aluminum hermetic lid with pinhole on the pan.
c. Seal the pan and lid using an encapsulating press.
d. Heat at a rate of 20 °C/min to a temperature of 300°C.
e. Determine the melting point of the sample using the onset of the endothermic peak. Peak onset is the point of intersection of the tangent of the peak with the extrapolated baseline.

4.7.3 Insoluble particles. Weigh three samples of 17 + 0.1 grams each (dry weight) from the prepared HMX sample. Transfer each sample into a 1000 mL beaker and add 500 mL of acetone to each beaker. Heat the contents of the beakers until the explosives are in solution. Combine the solutions and filter through USSS 40 and USSS 60 sieves. The filtrate shall be retained and used for testing acetone and inorganic insolubles. The screens shall be dried. Any particles remaining on the screens will be counted and reported as insoluble particles.

4.7.4 Acetone insoluble. Use the residue and filtrate from 4.7.3 or weigh an approximately 10 gram portion of the dried sample accurately to the nearest 0.01 gram, place the sample in a 600 mL beaker and add 400 mL of filtered acetone. Place the beaker and contents on a steam bath (cover with a watch glass if necessary) and stir occasionally until all of the HMX has dissolved. Filter the HMX solution through a 25 mL medium porosity porcelain filtering crucible, which has been ignited at 700 ± 20 degrees C, and tared to the nearest 0.1 milligram.

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Transfer any remaining insoluble material from the beaker to the crucible with the aid of a stream of acetone from a wash bottle. Wash the material remaining in the crucible three times with 20 mL portions of acetone and aspirate until the odor of acetone is no longer noticeable. Dry the crucible in an oven at 105 ± 5 degrees C for thirty minutes, cool in a desiccator and weight to the nearest 0.1 milligram. Reserve the crucible and contents for determination of inorganic insoluble materials given in 4.7.5.

Calculate the acetone insoluble material as follows;

$$\text{Percent acetone insoluble material} = \frac{(A - B) \times 100}{W}$$

Where:

A = The final weights of the acetone insoluble material and the crucible in grams
B = The tare weight of the crucible in grams
W = The HMX sample weight in grams.

4.7.5 Inorganic insoluble material. Ignite the sample reserved from the acetone insoluble material determination, 4.7.4, in a muffle furnace at 700 ± 10 degrees C for 30 ± 5 minutes, cool in a desiccator and weigh to the nearest 0.1 milligram.

Calculate the inorganic insoluble material as follows;

$$\text{Percent inorganic insoluble material} = \frac{(C - B) \times 100}{W}$$

Where:

C = The weight of the crucible and contents after ignition in g.
B and W are as given in 4.7.4.

4.7.6 Acidity. Weigh an approximate 10 gram portion of the dried sample accurately to the nearest 0.01 gram, place in a clean 800 mL beaker and add 500 mL of acetone. Place the beaker on a steam bath and heat with occasional stirring until the sample has completely dissolved. One hundred mL of distilled water shall be added and the mixture shall be cooled and titrated with 0.05 normal (N) sodium hydroxide using phenolphthalein or methyl red indicator. A blank shall be run and the results of the titration of the specimen shall be corrected for acidity of reagents. The acidity shall be calculated on the dry basis to percent acetic acid as follows:

$$\text{Percent acid (as acetic acid)} = \frac{6.0 \times (S - B) \times N}{W}$$

Where:

S = mL of sodium hydroxide solution used in sample.
B = mL of sodium hydroxide solution used in blank.
N = normality of sodium hydroxide solution.
W = weight of specimen on a dry basis, in grams.

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4.7.7 Impact sensitivity.

4.7.7.1 Safety. Proper safety should be exercised in determining the impact sensitivity of HMX. Compliance with local safety rules is required. The following safety precautions apply to both the ERL (4.7.7.3) and the Holston (4.7.7.4) impact sensitivity test methods.

Observe normal safety precautions -- pay particular attention to the following rules:

a. Operate the exhaust fan while tests are being performed.
b. Always operate the machine with a safety shield in place.
c. Secure the weight with a safety chain while cleaning the machine.
d. Remove waste explosives from the room after completion of each test.

4.7.7.2 Interferences. The following interferences apply to both the ERL (4.7.7.3) and the Holston (4.7.7.4) impact sensitivity test methods.

a. A machine loosely assembled or not in proper alignment may produce incorrect values.
b. A rough surface or cracks on the anvil or plunger may produce lower sensitivity values.
c. Insufficient or unevenly distribute sample may produce incorrect values.
d. A sample containing glass, metal or other gritty matter foreign to the product may produce low sensitivity values. (A low sensitivity value indicates more sensitive material.)
e. Wet samples or samples containing oil, greases and soft plastics may produce high sensitivity values.

4.7.7.3 ERL, Type 12, 2.5 kg impact sensitivity method. The impact sensitivity of HMX shall be determined using the ERL type 12 impact machine with a 2.5 kg drop weight on a 35 mg sample.

The following ERL test method is described in AOP-7:

a. Sample preparation. Pass a dry sample of HMX through a sample splitter until a representative portion of approximately 10 gram is obtained. A small scoop will be used to measure the HMX powder sample. Weigh a few scoops on a balance to determine the volume necessary to obtain 35 mg. Subsequent 35 mg samples shall be measure on a volume basis using the small scoop. The sample shall be poured directly onto a 180A garnet sand paper (1" x 1" square) with the sand side up resting on flat surface. Approximately 25 samples of 35 mg HMX each on a one inch square sand paper shall be prepared. Each of the samples shall be weighed on the garnet paper, (any abnormally high or low weigh should be adjusted), and placed on the anvil for testing.
b. Performance of the test. Test data shall be recorded on a worksheet which can be on paper (hard copy) or on a computer. Record sample name, ID number or lot number, sample preparation (powder, cube, etc.), requester, operator, date, temperature, and humidity. The drop heights used for the test are as follows:

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64.0 cm
50.5 cm
40.5 cm
32.0 cm
25.5 cm
20.0 cm
16.0 cm
12.5 cm
10.0 cm
8.0 cm
6.5 cm
5.0 cm

NOTE: The actual starting height for HMX should be determined from test data on HMX previous tested.

Before each drop, the striker and anvil shall be cleaned. Ethanol or acetone on a wipe shall be used to clean the surfaces. A spatula or sandpaper may also be used to clear any further residue from previous tests. A 35 mg sample (on the sand paper) shall be placed on the anvil. Gently place the striker on top of the sample. Close the shield. Turn on the electromagnet, and using the hoist, raise the weight to the desired height. Face the opposite direction of the impact machine and turn the electromagnet switch to the "off" position, allowing the weight to fall and strike the top of striker. Lift and secure the weight, remove the striker to determine whether the drop was a positive (explosion) or negative (nonexplosion) reaction. Any reaction observed is considered a go: smoke, flame, sparks, odor, burn marks on the striker or anvil, a noise above the ambient drop weight noise. A positive reaction shall be recorded as "1", and a negative reaction as a "0". When a positive reaction is achieved, the next drop height shall be one level lower. When a negative reaction is achieved, the next drop height shall be one level higher.

NOTE: The actual test begins when a positive reaction is followed by a negative reaction, or a negative reaction is followed by a positive reaction. It may be necessary to make several test shots before the starting point of the test is recorded.

Repeat 4.7.7.3 (b) until all 25 samples prepared in 4.7.7.3 (a) are drop-tested and recorded.

c. Test result calculations. The test result calculation may be preprogrammed in a computer and automatically calculated based on the results recorded in the computer. If the test was recorded on paper, enter all data into the impact spreadsheet. In either case, a hard copy of the results shall be immediately printed following the testing for record keeping. The following is an explanation of the calculation for the height in centimeters where 50 percent of the explosions occur:

The lowest drop height with recorded data is assigned an "I" value of 0. Each successively greater height is assigned values of 1, 2, etc. The following values are calculated:

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ni = # of negatives at each log height.
N = sum of ni
A = sum of (i*ni)
B = Sum of i*i*ni
C = log interval of lowest drop height
M = C + 0.1(A/N + 0.5)
Height (at 50% explosion) = 10 raised to the power of M.
S (standard deviation, log units) = 0.162*[(NB - A*A)/N*N + 0.0219]

The ERL, type 12, 2.5 kg impact machine shall be tested for accuracy each day that it is used for testing HMX in accordance with the specification. For this purpose an HMX sample having a known sensitivity range shall be used. The results shall be recorded on a chart showing the minimum and maximum values for the sample (control limits) and the midpoint of the control limits range. If the test result for the control sample falls outside or near the minimum or maximum value of the control limits, the impact machine shall be checked and adjusted if necessary to assure that the platform for the anvil is level and the guide rails for the drop weight is upright. Drop height is also checked periodically with vertical weighed line to ensure that the impact machine is upright. Drop tests with carbon paper for determination of points of impact shall be conducted, and impact machine shall be adjusted as required.

4.7.7.4 Holston impact sensitivity. The Holston impact sensitivity of HMX shall be determined using a Holston impact machine.

4.7.7.4.1 Holston short impact method. Pass a dry sample of HMX through a sample splitter or roll and quarter on glazed paper until a representative sample of about 5 grams is obtained. Position 13 brass percussion caps (0.303 inch diameter and 0.20 inch in height) with open end up on a flat surface. Fill a 0.025 g loading spoon with HMX, remove the excess material by passing the flat edge of a wooden spatula across flat surface of the spoon and transfer the portion remaining in the spoon to one of the percussion caps. Repeat the HMX loading procedure until all of the percussion caps are loaded. Be sure the HMX is evenly distributed in each of the percussion caps. Turn on the exhaust fan in order to remove the dust and fumes from the impact machine enclosure. Using laboratory tongs place a loaded percussion cap on the anvil of the impact machine. Insert a plunger through the guide hole above the anvil and into the open end of the percussion cap. Turn the electromagnet power switch on. Turn the gear handle until the base of the lower electromagnet arm coincides with the 26 cm mark on the right guide bar. Lower the safety shield and lift the weight vertically until it is held in place by the electromagnet. Face the opposite direction of the impact machine and turn the electromagnet switch to the OFF position allowing the weight to fall and strike the plunger. Lift the weight and raise the safety shield and allow the weight to rest on the safety shield. Examine the percussion cap to determine if an explosion has occurred. If an explosion has occurred, the percussion cap is usually disintegrated. A partial explosion may be recognized by inspection of the cap for parts of the rim having been blown away. An explosion may also be recognized by a sharp report or smoke in the area of the plunger. Clean away all of the unexploded material and percussion cap parts from the anvil, plunger and base plate with a brush or cloth prior to placing the next test sample in place.

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a. If the above test is a non-explosion, place a minus sign on line zero (0) above Test No. 1 on the short impact method test card. (See below for a sample test card.) Repeat the procedure with one of the loaded percussion caps, and continue testing on line zero until an explosion occurs or the acceptance (A) zone has been reached. If an explosion occurs, move up to the next line on the card and continue testing until an acceptance or rejection is determined.

b. If the above test was an explosion, place a plus sign on Line 1 above Test No. 1. Should a second explosion occur, record the plus sign on Line 2 above Test No. 2. The sample would be rejected and a long impact must be performed. If the second sample is not an explosion, continue testing and record explosions or non-explosions on their appropriate line until acceptance or rejection is determined.

c. If the sample being tested is rejected, a long impact value must be determined in accordance with 4.7.7.4.2. Otherwise (if the sample passes the short impact test) report sample as accepted on impact.

Sample test card for Holston short impact test method

SHORT METHOD
IMPACT SENSITIVITY

RUN HMX AT 26 CM
RUN RDX AT 33 CM

IF BATCH IS REJECT – RUN IMPACT BY LONG METHOD USE OTHER SIDE OF CARD

ACCEPT_____
REJECT_____
LONG METHOD_____CM

	4								R	R	R	R	R	R
NUMBER	3			R	R	R	R	R	o	o	o	o	o	A
OF	2		R	o	o	o	o	o	o	o	o	o	A	A
EXPLOSIONS	1	o	o	o	o	o	o	o	o	o	o	A	A	A
	0	o	o	o	o	o	A	A	A	A	A	A	A	A
		1	2	3	4	5	6	7	8	9	10	11	12	13

Test Number

Note: A – Zone for acceptance
R – Rejected by the short method, and a test with the long impact method (see 4.7.7.4.2) must be performed.

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4.7.7.4.2 Holston long impact method. Pass a dry sample of HMX through a sample splitter or roll and quarter on glazed paper until a representative sample of about 5 gram is obtained. Position 25 brass percussion caps (0.303 inch diameter and 0.20 inch in height) with open end up on a flat surface. Fill a 0.025 g (25 mg) loading spoon with HMX, remove the excess material by passing the flat edge of a wooden spatula across flat surface of the spoon and transfer the portion remaining in the spoon to one of the percussion caps. Repeat the HMX loading procedure until all of the percussion caps are loaded. Be sure the HMX is evenly distributed in each of the percussion caps. Turn on the exhaust fan in order to remove the dust and fumes from the impact machine enclosure. Using laboratory tongs place a loaded percussion cap on the anvil of the impact machine. Insert a plunger through the guide hole above the anvil and into the open end of the percussion cap. Turn the electromagnet power switch on. Turn the gear handle until the base of the lower electromagnet arm coincides with the 30 cm mark on the right guide bar. Lower the safety shield and lift the weight vertically until it is held in place by the electromagnet. Face the opposite direction of the impact machine and turn the electromagnet switch to the OFF position allowing the weight to fall and strike the plunger. Lift the weight and raise the safety shield and allow the weight to rest on the safety shield. Examine the percussion cap to determine if an explosion has occurred. If an explosion has occurred, the percussion cap is usually disintegrated. A partial explosion may be recognized by inspection of the cap for parts of the rim having been blown away. An explosion may also be recognized by a sharp report or smoke in the area of the plunger. Clean away all of the unexploded material and percussion cap parts from the anvil, plunger and base plate with a brush or cloth prior to placing the next test sample in place. Repeat the procedure with the remaining HMX loaded percussion caps until twenty trials have been recorded. Raise the electromagnet 5 cm after each non-explosion and lowering the electromagnet 5 cm after each explosion. The first non-explosion after an explosion shall be considered as the starting point of the twenty trials. Record the electromagnet height for this trial in cm as the first of the twenty trials. Place a minus sign (-) opposite the height in cm indicating the trial resulted in a non-explosion. Record the electromagnet height for each subsequent trials and either a minus sign to indicate a non-explosion or a plus sign (+) to indicate an explosion.

Calculate the percentage explosions at a given height using the following equation:

$$\text{Explosions, 50\% point} = \frac{A \times 100}{B}$$

Where:

A = The number of explosions at a given height.
B = The total number of explosions and non-explosions at the given height.

Calculate the impact sensitivity as follows:

$$\text{Impact sensitivity, cm} = C - \frac{5 \times (D - 50)}{D - E}$$

Where:

C = The lowest height in cm at which more than 50% explosions occurred.
D = The percentage of explosions greater than 50%

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E = The percentage of explosions less than 50%.
5 = The difference in height in cm of each test.

The Holston impact machine shall be tested for accuracy each day that it is used for testing HMX in accordance with the specification. For this purpose an HMX sample having a known sensitivity range shall be used. The test shall be run in accordance with 4.7.7.4.2. The results shall be recorded on a chart and shall fall within the minimum and maximum values for the control sample (control limits).

4.7.8 Granulation. The granulation shall be determined by the following procedure:

A 50 ± 0.05 gram sample of HMX (dry weight), weighed accurately to the nearest 0.01 gram, shall be transferred to a 600 mL beaker containing approximately 300 mL of a 2 percent solution of a suitable wetting agent such as dioctyl sodium sulfosuccinate (see 6.5). With the aid of a rubber policeman attached to a glass stirring rod, the mixture shall be stirred for a few minutes wetting the sample thoroughly and breaking up as many of the aggregate as possible.

A spray nozzle (see 6.6.2) under tap water pressure, shall be used to quantitatively transfer this mixture to the uppermost sieve of a set of the specified 8-inch U.S. Standard sieves complying with ASTM E11, nested in order of decreasing size, the largest mesh being placed on top. This assembly should be set up near a water tap and drain. Provisions shall be made to prevent transfer of the explosive to the drain. The pressure of the spray shall be adjusted so that when the spray strikes the sample at an angle approximately perpendicular to the screen from a height of 2 to 3 inches, it is possible to wash the sample back and forth across the sieve without splashing any of the material over the side of the sieve. The spray should be moved about the screen at a rate such that the spray would traverse the diameter of the screen 1 to 2 times per second. The wet agglomerates shall be gently crushed on the top sieve with the aid of a rubber policeman and the washing of the material back and forth across the sieve with the water spray shall be continued until all the agglomerates have been broken and only individual crystals larger than the mesh of the sieve remain on the sieve. The rubber policeman shall be used in breaking up the agglomerates on only the uppermost sieve. The top sieve shall be removed and a few drops of a 10 percent solution of the wetting agent shall be added to the material on the next sieve and washed with the water spray as before for a maximum of 2 minutes or until no change is noticed in the amount of crystals remaining. This procedure shall be repeated for each of the sieves.

After the washing has been completed, the portions remaining on each of the screens shall be quantitatively transferred by means of a suitable transfer funnel into a separate previously tared fritted glass filtering crucibles of coarse porosity as follows:

The screen shall be held in an almost vertical position and with a moderate spray of water from the spray nozzle, the material shall be gently washed to the lower part of the screen by drawing the spray back and forth across the screen, beginning at the top and moving slowly down the screen as the crystals move down. When the material has been collected at the lower part of the sieve, it can readily be washed into the crucible with a stream of water from a wash bottle.

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The crucible shall be aspirated during the transfer process and the crucible and contents shall be aspirated for approximately 2 minutes after the transfer has been completed. The suction shall be turned off and 15 mL of anhydrous methanol added to the crucible and contents; the methanol shall be allowed to remain in contact with the explosive for approximately 5 seconds and then removed with the aid of suction. The contents of the crucible shall be washed once more with anhydrous methanol as described above and then the crucible and contents shall be aspirated until the odor of methanol is no longer discernible. The crucible and contents shall be dried in an oven maintained at 105 ± 2 degrees C for 15 minutes (or equivalent method), cooled in a desiccator and weighed.

The weight of the material retained on each sieve shall be determined and the percentage through each sieve calculated on the basis of the dry weight of the HMX sample to determine compliance with 3.2.1 (Table II).

4.7.9 Workmanship. Visually observe the production of HMX and examine the samples submitted for testing for compliance with the requirements given in 3.4.

Attribute sample inspection shall be conducted in accordance to Table III (see 4.5.2) using the inspection levels in Table III as stated in the classification of characteristics paragraphs below (4.7.9.1 to 4.7.9.5).

NOTE: See MIL-STD-1916 for definitions of critical, major and minor classification of characteristics.

4.7.9.1 Classification of characteristics – loaded drum with tied bag, cloth (see 5.1 and 6.4.1).

Classification	Examination/test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Foreign matter	Level II	3.4	Visual
102	Bag pierced or torn	Level II	3.4	Visual
103	Not tied or improperly tied	Level II	3.4	Visual
Minor	None defined			

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4.7.9.2 Classification of characteristics – empty polyethylene bag (see 5.1 and 6.4.1).

Classification	Examination/test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Seam splits when pressure is applied along entire length of seam	Level II	3.4	Visual/Manual
102	Bag damaged	Level II	3.4	Visual
Minor	None defined			

4.7.9.3 Classification of characteristics – loaded drum with tied polyethylene bag or sealed rubber bag (see 5.1 and 6.4.1).

Classification	Examination/test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Bag improperly closed	Level II	3.4	Visual
102	Bag damaged	Level II	3.4	* See Note
103	Insufficient solution	Level II	3.4	* See Note
104	Insufficient alcohol in solution	Level II	3.4	* See Note
105	Net weight	Level II	3.4	* See Note
Minor	None defined			

Note: *Inspection will be accomplished by witnessing procedures used by contractor to comply with requirements.

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4.7.9.4 Classification of characteristics – loaded drum with cover and locking ring before closing (see 5.1 and 6.4.2).

Classification	Examination/test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Top chime bent, deformed, or cut	Level II	3.4	Visual
102	Bottom chimes collapsed (annular groove closed or partially closed), or deformed	Level II	3.4	Visual
103	Body bulged, cut or dented	Level II	3.4	Visual
104	Gasket in cover missing or damaged	Level II	3.4	Visual
105	Cover bent, creased or deformed in gasket area or around edge	Level II	3.4	Visual
106	Locking ring damaged so as to prevent closing	Level II	3.4	Visual/manual
Minor				
201	Outer body surface seriously scuffed or metal scratched through galvanized surface	Level II	3.4	Visual
202	Nicks or dents in chimes or cover not affecting function	Level II	3.4	Visual
203	Locking ring bent or deformed	Level II	3.4	Visual

4.7.9.5 Classification of characteristics – sealed drum (see 5.1 and 6.4.2).

Classification	Examination/test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Alcohol solution leaking from drum	Level II	3.4	Visual/manual
102	Cover or locking band improperly seated	Level II	3.4	Visual
103	Seal missing or improperly applied	Level II	3.4	Visual
Minor				
201	Marking missing, incorrect or illegible	Level II	3.4	Visual

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5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in contract or order (see 6.2 and 6.4).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. HMX is to be used as an energetic ingredient in a number of explosive compositions for applications where high energy is needed in ammunition items.

6.2 Acquisition requirements. The acquisition document should specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
c. Provisions for submission/resubmission of first article samples. A first article quantity, or portions thereof, as directed by the Contracting officer in accordance with contract provision: Instruction Regarding Submission of First Article, should also be submitted whenever there is a lapse in HMX production of a period in excess of 90 days, or whenever a change occurs in the manufacturing process, or in sources of constituents or process materials, drawings or specifications which could affect product uniformity as determined by the Government.
d. Acceptance and description sheet – Acceptance and description sheets should be prepared for each lot in accordance with MIL-STD-1171.
e. Quantity required and delivery schedules.
f. Grade and granulation class required (see 1.1 and 3.2).
g. Levels of preservation and packing (see 5. and 6.4).
h. Acceptance and description sheets (see 6.3).
i. Item hazard classification (see 6.4.5).

6.3 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract. In addition, the following data requirements should be considered when this specification is applied on a contract, per the Contract Data Requirements List, DD Form 1423:

Reference	DID Number	DID Title	Suggested Tailoring
6.4.4	DI-PACK-81059	POP Test Report	None

6.4 Acceptable packaging requirements. The following packaging and marking requirements have been used for packaging HMX, and is found to be acceptable to the Government. These requirements should be included in the contract or order for the

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procurement of HMX because HMX is a hazardous energetic material. (Caution: If the following paragraphs are to be incorporated in a contract, they should be modified, using standard contract language, to make them compulsory requirements.)

6.4.1 Preservation and packaging.

6.4.1.1 Level B and Level C. Unless otherwise specified by the procuring activity, HMX will be preserved and packaged as follows: Not more than 50 pounds dry weight, of the wet HMX should be packed in a cloth bag described in 6.4.1.1.2. Filled cloth bags should be placed in a rubber bag, rubberized cloth bag or two polyethylene bags, described in 6.4.1.1.1, placed one inside the other. The bags should be free of foreign matter. A water-isopropyl alcohol solution in a 40 percent isopropyl alcohol and 60% water ratio should be added to the outer bag. The amount of water added has to comply with DOT requirements (49 CFR 100-199). Inner and outer bags should be securely tied, with the polyethylene bags individually tied, using non-metallic tape or cord. The tops of the outer bags should be gathered and formed into a gooseneck when being tied.

6.4.1.1.1 Polyethylene bags. The polyethylene bags should comply with Type II Style 1 of Specification PPP-B-26 except that closure will not be heat sealed. The bag size should be approximately 31 inches flat width by 55 inches in length. The bag seams should meet the seam strength test with an AQL of 0.65 percent at 50 percent of the breaking strength of the polyethylene film. The seam strength test should be performed in accordance with Method A of ASTM D882 using one-inch wide specimens. Seams should also be examined by separating the bag face and applying pressure manually along the entire length of the seam. Seams which can be opened at any point in this manner are not acceptable.

6.4.1.1.2 Cloth bags. The cloth bags should be made from white cotton twill, free of size and brighteners, complying with the general requirements of Specification CCC-C-461 and the following detailed requirements:

Yarn	Carded
Yarn Ply	Warp 12/1 Fill 18/1
Weight	7.2 + 0.5 oz/sq yd (Greige) 6.0 + 0.7 oz/sq yd (after bleaching)
Yards per inch (minimum)	Warp 78 Fill 47
Breaking strength	Warp 170 Fill 70
Weave	2 x 1 twill

A suggested bag size is 19 inches wide by 29 inches in depth. The tie tape may be attached to the bag by sewing. Alternatively, double filled gray cloth cotton duck weighing not less than 12 oz. per square yard may be used for the bag.

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6.4.2 Packing.

6.4.2.1 Level B. A maximum of 300 pounds dry weight of crystalline HMX packaged in accordance with 6.4.1 should be packed in an outer bag which has been placed in a drum complying with Department of Transportation Specification 1A2, Paragraph 178.504 of the Code of Federal Regulation 49 CFR 100-199.

6.4.2.2 Level C. A maximum of 225 pounds dry weight of crystalline HMX packaged in accordance with 6.4.1 should be packed in a fiber drum complying with DOT specification 1G, Paragraph 178.508 of the Code of Federal Regulation 49 CFR 100-199 (approximate gross weight of 300 pounds) and the following requirements. The drum should be approximately 18 1/2 inches in diameter by 26 inches in height, inside dimensions. The drum should have a 24-gauge steel cover with a rubber gasket, level locking band with provisions for sealing wire and wide bottom chime (2-inch minimum formed height). All metal parts should be hot dip galvanized and chimes should be welded. Bottom chimes should be 0.028 inch steel and top chimes should be 0.025 inch steel. The body should be wound with hot melt or thermoplastic adhesive and the seam covered with a hot sealed tape. The inner surface of the bottom and body should have a laminated liner of 0.002 inch thick polyethylene. The bottom crimp has to be caulked. The finished drum including closure should be leak tight construction suitable for liquid contents.

6.4.3 Marking. Containers should be marked as required by the Code of Federal Regulation 49 CFR 100-199. In addition, shipments should be marked in accordance with Drawing 12982865. Each container should be clearly labeled with the lot number and net weight of its contents. The HMX contents of the container should be from the same lot.

6.4.4 Performance oriented packaging (POP). The exterior pack cited above should meet all of the POP test requirements in accordance with the Code of Federal Regulations, Title 49 (49CFR), including testing, and retesting as specified in Paragraph 178.601 (e) of 49 CFR. A POP test report should be generated in accordance with DI-PACK-81059 following the tests. POP test report may be waived if an acceptable non-government analogy can be made IAW 49CFR to another pack which has successfully completed the testing. This analogy should also be documented IAW DI-PACK-81059. When completed, either POP test report should be kept on file by the contractor and should also be submitted to the U.S. Army Research Development and Engineering Center, ATTN: AMSTA-AR-AEP, Picatinny Arsenal, New Jersey, 07806-5000. (NOTE: If a POP test report is prepared against an acceptable analogy, the analogy POP test report should also be submitted to AMSTA-AR-AEP). The POP marking symbol applied to the exterior pack should be that belonging to the organization which conducted the POP testing.

6.4.5 Item hazard classification. All U.S. manufacturers should make certain that the item is tested in accordance with Part 173, Subpart C, Section 173.58 (a) of 49 CFR, Parts 106-180 to assign proper Class and Division for all explosives (division 1.1, 1.2, 1.3 and 1.4 explosives.) Registration with the Associate Administrator of Hazardous Materials safety is required in accordance with Part 173, Subpart C, Section 173.56 (b) (l) or 173.56 (c) of 49 CFR so that proper markings in accordance with Part 172, Subpart D, Section 172.301 (a) and 172.320 (a) are met.

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All foreign manufacturers should make certain that the dangerous goods are tested in accordance with United Nations Committee of Experts on the Transportation of Dangerous Goods (as published in UN Document ST/SG/AC. 10.11 latest revision, Recommendations for the Transport of Dangerous Goods - Tests and Criteria) to determine the proper class and division (Class 1-9 and Division 1.1 -1.6 for explosives). Registration for air and vessel transport is required with each manufacturing country's National Competent Authority. The Hazard Classification letter of competent authority is issued in accordance with part 2, paragraph 1.3 of the International Civil Aviation Organization (ICAO) technical instructions and approves the hazard classification and compatibility group assignment and assigns the appropriate shipping name to the dangerous goods. The proper packaging, marking and labeling is contained in the United Nations Committee of Experts on the Transport of Dangerous Goods (as published in UN Document ST/SG/AC. 10.1, latest revision, recommendations on the Transport of Dangerous Goods).

For air transport the dangerous goods should comply with the provisions of the International Air Transport Associate (IATA) Dangerous Goods Regulations and for vessel transport, the dangerous goods should comply with the provisions of the Intergovernmental Maritime Organization's International Maritime Dangerous Goods Code (IMDG Code).

6.4.6 Referenced documents for packaging. The following list of documents referenced in 6.4 should be included in the contract or purchase order as requirement documents. Document users are cautioned that they should meet all requirements of these documents if cited in the contract or purchase order.

a. Federal and military specifications/standards. (See 2.2.1 for issues of documents and address to obtain these documents.)

CCC-C-461 – Cloth, Twill, Uniform Cotton
PPP-B-26 – Bag, Plastic, Polyethylene

b. Other Government documents, drawings and publications. Unless otherwise specified, the issues of documents are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 – Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite, "49 CFR 100-199 (latest revision)".)

c. Non-government publications. The issues of the following documents which are DOD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise

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specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation. (See 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation D882 – Tensile Properties of Thin Plastic Sheeting

(Applications for copies should be addressed to the American Society for Testing Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959)

6.5 Dioctyl sodium sulfosuccinate. Dioctyl sodium sulfosuccinate is commercially available as Aerosol OT.

6.6 Approved equipment.

6.6.1 MicroBondapak (CN) HPLC column. The MicroBondapak CN column (Catalog No. WAT086688) is available from the Millipore Corporation, Waters Chromatography Division, headquartered in Milford, MA.

6.6.2 Spray nozzle. A spray nozzle can be made by fitting a Gooch crucible (Number 5, Coors porcelain crucible containing about 75 openings approximately 0.07 centimeters (cm) in diameter has been found satisfactory) over a Number 8 one hole rubber stopper fitted with a short piece of glass tubing to which is attached a length of rubber hose approximately 1 cm inside diameter. The free end of the hose is connected to the water tap and the Gooch crucible at the other end acts as a spray nozzle.

6.7 Virgin HMX. Virgin HMX is produced with a feedstock to the recrystallizer that is solely composed of slurried HMX coming directly from the HMX synthesis process.

6.8 High purity HMX (beta), RDX calibration standards. High purity HMX (beta) and RDX calibration standards can be obtained from the following addressees:

a. Commander, Holston Army Ammunition Plant, 4509 West Stone Dr., Kingsport, TN 37660-9982.
b. Commander, ARDEC, ATTN: AMSTA-AR-AEE-W, Picatinny Arsenal, NJ 07806-5000.
c. Naval Explosive Ordnance, Technology Division, Bldg. 2172, ATTN: Mr. Lou Wasserzug (Code 50A24), 2008 Stampneck Rd., Indian Head, MD 20640.

6.9 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000.

6.10 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be

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submitted through the Contracting Officer to Commander, ATTN: AMSTA-AR-QAT-P, ARDEC, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.11 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification. (See 6.9).

6.12 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement NATO STANAG 4284. When amendment, revision or cancellation of this specification is performed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.13 Subject term (key word) listing.
Explosive compositions
Her majesty explosive
High melting explosive

6.14 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Custodians	Preparing activity:
Army - AR	Army-AR
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